



SDEV2171

Lesson 03: Creating Your First Expo App

A LEADING POLYTECHNIC COMMITTED TO YOUR SUCCESS

Lesson Objectives

By the end of this lesson, students should be able to:

- Create a new React Native application using Expo tooling
- Run an Expo application on an emulator or physical device
- Identify key files and folders in a default Expo project
- Modify existing application code and observe changes in the running app
- Use basic troubleshooting techniques to identify and resolve common Expo startup or runtime issues during project creation

Lesson Roadmap

1. Lesson-02 status check and recovery path
2. Course template and version choice
3. Instructor project-creation demo
4. Guided build and first run
5. Project structure walkthrough
6. Small code changes and troubleshooting lab
7. Checkpoint and next lesson bridge

Starting States From Lesson 02

- `verified` : start project creation now
- `partial` : finish the recorded next action first, then create the app
- `blocked` : do not create a new project yet; get instructor approval first

Course rule

Do not hide blockers. Record them and recover from evidence.

Recovery Window And Re-entry Point

First 10 minutes

- `verified` : begin Step 2 immediately
- `partial` : complete the missing setup step first
- `blocked` : work with the instructor on one recovery action

Re-entry point

Join the main lesson as soon as you have:

- Node.js working at the course version
- All required setup steps complete (e.g., Java, Android Studio, a running emulator)
- one approved launch path

If you do not recover in time

- use the instructor reference project for the file-structure and code-edit parts
- leave class with a recorded next step for your own project



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Today's Build Target

- one new Expo app created from the terminal
- one working development-build launch path
- one visible edit in the running app
- one short explanation of the project structure

Why this matters

Lesson 04 builds on the same project instead of starting over.

Course Template Choice

Official Expo baseline

- Expo SDK 56
- React Native 0.85
- React 19.2.3
- Node.js 22.13.x or newer

Course choice for lesson 03

- use the `blank` template
- keep the project small enough to understand in one class
- add navigation later after students understand the app entry point

Exact Project-Creation Command

From the folder where you want the project to live, run:

```
npx create-expo-app@latest lesson03-first-app --template blank@sdk-57
cd lesson03-first-app
npx expo install expo-dev-client
npx expo run:android
npx expo start
```

Important version note

- Do not omit the `--template blank@sdk-56` part in this lesson
- `create-expo-app@latest` alone still creates an SDK 54 project during the current transition period
- after `expo-dev-client` is installed, `npx expo start` targets the development build automatically

Launch Paths For Today

Preferred classroom path

- Android emulator with a development build, or
- iOS simulator on macOS with a development build

Physical-device path

- use it only if the SDK 56 development-build workflow was already prepared in lesson 02
- if not, stay on emulator or simulator for lesson 03

Course rule

One working path is enough.

What To Watch During The Demo

- the project folder is created successfully
- dependencies install without stopping the workflow
- `expo-dev-client` installs
- the native development build installs
- the Expo server starts
- the app opens on one working path
- the first edit appears after saving the file

Workflow Map

Step	What you do	What you should see
1	Run the create command	New project folder appears
2	Install <code>expo-dev-client</code>	Project is ready for a dev build
3	Run <code>npx expo run:android</code> or <code>npx expo run:ios</code>	Development build installs
4	Run <code>npx expo start</code>	Expo server starts
5	Launch the app	Generated starter app opens
6	Open <code>App.js</code>	Main lesson UI file is visible
7	Save one edit	Running app updates

What Your Screen Should Look Like

- the newly generated starter app will be simpler than the instructor reference app
- your first success does not need to match the demo screen exactly
- the important proof is:
 - the project was created
 - the app opens
 - your saved edit appears

Why the early instructor app looks different

It is a prepared demo variant used to point out files, edits, and reload behaviour more clearly.

Key Project Files

- `App.js` : the main screen code for this lesson's generated starter app
- `app.json` : project configuration such as app name and slug
- `package.json` : dependencies and scripts
- `assets/` : images and static files
- `node_modules/` : installed packages, not a folder students edit directly

First question to ask

Where is the visible UI actually defined?

Simple folder map

```
lesson03-first-app/  
  App.js  
  app.json  
  package.json  
  assets/  
  node_modules/
```



Small Changes Students Should Make

- change the title text
- change one paragraph or instruction line
- add one small `View` or `Text` block
- change one color value

Why these edits matter

They connect file edits to visible UI changes without creating too much new code at once.



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Troubleshooting Lab

Common issues

- project creation command fails
- Expo server does not start
- device path does not open the app
- file changes do not appear

Response pattern

1. identify the failing step
2. try one small fix
3. recheck the result
4. record the evidence

Launch Actions After `npm run start`

- Android emulator: press `A` in the Expo terminal UI
- iOS simulator on macOS: press `I` in the Expo terminal UI
- prepared device path: follow the instructor's SDK 56 launch step

If nothing opens

- confirm the correct simulator or emulator is already running
- retry one launch action
- switch to another approved path if needed

Checkpoint

Students should be ready to show

- a running lesson-03 Expo app
- one visible code change in that app
- one key file or folder and its purpose

Students should also be able to answer

- Which command created your project?
- Which launch path are you using?
- What troubleshooting step did you use or would you use first?

What We Learned Today

- how to create a new Expo app on the course SDK 56 baseline
- how to start the app and verify one working run path
- where the main files in a blank Expo project live
- how saving code changes affects the running app
- how to troubleshoot creation, startup, connection, and reload problems step by step

Next Lesson Bridge

- lesson 04 uses this same app
- next class focuses on core React Native components
- keep today's project available and do not replace it with a new template
- preserve a known-good `App.js` state that still runs

Ready for lesson 04 means

- your project folder still opens
- the app still runs
- one visible edit remains in place
- you can point to `App.js` as the file where lesson 04 work will begin